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Smart Well Design aligned with Geological Objectives to Deliver Value and Cost Efficiency

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Abstract

A good well design is essential for any drilling project where the primary objective is to deliver value for the entire life-cycle of the well. It becomes imperative that Geological objectives are considered in the design phase to ensure that targets will be achieved optimally with minimum challenges and the lowest financial impact. This paper describes an approach followed by the drilling team to design a deep gas well to maximize the efficiency of drilling operations.

The drilling team analyzed the short-term and long-term objectives of the well and utilizing in-house developed software tools addressed all the associated challenges. The biggest challenge facing the team was a tight, flow-capable zone, which would be combined with the reservoir section in the same hole. Through the use of heat maps, flow risk was determined, and an appropriate drilling fluid density was selected. Moreover, drilling several zones of the same size without running a liner provided flexibility in terms of side-tracking and yielded immense cost savings.

Deviating from a typical well profile, the team developed a design that offered flexibility while maintaining cost efficiency. The main requirement from the Geological team was to drill a pilot hole and then decide either to do a small nudge away from the mother-bore, termed ‘twinning the well’, or to perform a full side-track away from the mother-bore based on log results. Drilling of this well commenced with a typical slim-bore design, with one exception that two known reservoir intervals were combined in one section without running a 7-in liner that separates both. The liner would have to be milled in order to side-track the well, which would have a huge time and cost impact; however, with optimized engineering design, this was not required. This section also contained the unpredictable tight flow zone that could jeopardize the project if encountered. However, with a thorough risk assessment and control measures in place, drilling operations were concluded successfully. Well was side-tracked from the open hole after placing two cement plugs and using open hole whip-stock, and in the subsequent lateral, the two reservoir sections were isolated from each other with a 7-in liner, which also covered a high-pressure water layer just above the production interval.

State-of-the-art data analysis and visualization tools helped to identify and mitigate risks associated with combining two reservoir sections, which also contained a tight, unpredictable flow zone. In addition, with a novel approach to well design, the requirement of milling a 7-in liner in order to side-track the well was

eliminated, which led to significant time and cost savings. All the Geological objectives were achieved successfully, thereby validating the smart approach to well design.

Background

Drilling deep onshore delineation gas wells requires thorough planning and preparation in order to generate an optimized well design. These wells have an appraisal characteristic and are essential to evaluate the reservoir fully before committing to completing the well. Typically, a pilot hole has to be drilled into the target reservoir layers to evaluate the presence of hydrocarbons and to ascertain reservoir properties such as porosity, water saturation, etc. Acquiring formation pressures and formation samples on wireline conveyed tools is common, and this information yields further information on reservoir permeability, mobility, pressures, and fluid content. These logs also help to differentiate between water and gas-bearing formations. Based on this evaluation, it may be decided to sidetrack the well to isolate water-bearing formations with an additional casing string. In some cases, twinning may be performed to move just a nudge away from the mother bore and re-drill the reservoir section. The water zone is typically present just above the reservoir and is isolated with a production liner. Drilling then continues into the target reservoir up to a depth that avoids penetrating the non-prolific reservoirs, which have been identified on the logs.

However, this twinning or side-tracking option comes at a cost. Due to the fact that in the pilot hole a production liner has been set at the top of the cap rock and since considerable focus is required regarding the productive interval of the reservoir, it is not possible to deviate from pilot hole and re-drill the reservoir in the available open hole section below the liner shoe. As a result, the production liner has to be milled to gain access to the open hole below the production casing and have enough footage available to nudge away and re-drill a lateral. The production liner is once again set on top of the cap rock in such a way that it isolates the water zone identified on the logs. The geological objectives cannot be met unless the liner is milled.

Milling a cemented liner is a very time and cost consuming activity. The milling process requires several trips to mill the liner hanger followed by the liner itself in preparation of the hole for a side-track. In particular the body of the liner hanger is difficult to mill and progresses at a very slow pace. Together this activity can consume weeks of rig time and the associated services and material costs could be extremely high. Any option to avoid this milling must be utilized if performance optimization is the eventual goal.

In order to enhance the well construction process, the drilling and reservoir teams need to evaluate the geological objectives closely and identify opportunities for improvement. In the subject well instance, the teams came together to come up with a novel solution that eliminated the need to run a production liner. It required combining a high-pressure flow zone in the same hole section as the reservoir formations in a size larger than normally drilled. In this way, after open hole logging is done, the same hole would be side-tracked from a shallower depth and a production liner set below the water zone to effectively isolate it. Then, drilling would continue into the reservoir in a smaller hole, which is typical of this field. Once drilled to the desired depth, the well would be cased off with a liner run as a lower completion and then completed with a production packer in the upper completion.

However, this strategy entailed certain challenges. The first and foremost of which was the presence of a potential water flow zone above the reservoir. Drilling through this zone is challenging because of its unpredictable nature. Due to its tight and non-homogenous character, it is not possible to identify if this zone will flow and required to be killed with a high mud weight. Secondly, if killed with a high mud weight, the hydrostatic pressure imposed on the weaker formations leads to severe loss circulation issues, which can have huge time and cost implications. Therefore, these formations would need to be isolated before drilling can progress, in which case the improved well design would revert to a typical well design in which a production liner is run to isolate the flow zone from the thief zones. A detailed analysis is required to be done on offset wells to ascertain the risk of influx.

Enabling Technology

To overcome the challenge of unpredictable flow, a smart web-based tool was developed that assists Drilling Engineers in assessing the risk of well control and suggesting a suitable drilling fluid density to avoid losses while over-balancing the formation. This python- based tool analyses the offset wells data in the near vicinity of the subject well, takes in to consideration the formation pressures encountered (typically taken as initial shut-in drill pipe pressures), transposes and correlates those pressure to the well under consideration, and displays the output as an interactive heat map of the field. This map further displays the output in color gradients of flow risk, as shown in Figure 1 below. Another very important result is the suggested mud weight, which utilizes the algorithm described above for calculation. Moreover, offset wells are plotted to indicate their proximity to the subject well. As a result, drilling engineers can identify the need for suitable technology to be deployed, such as Managed Pressure Drilling (MPD), in case the computed risk is high.

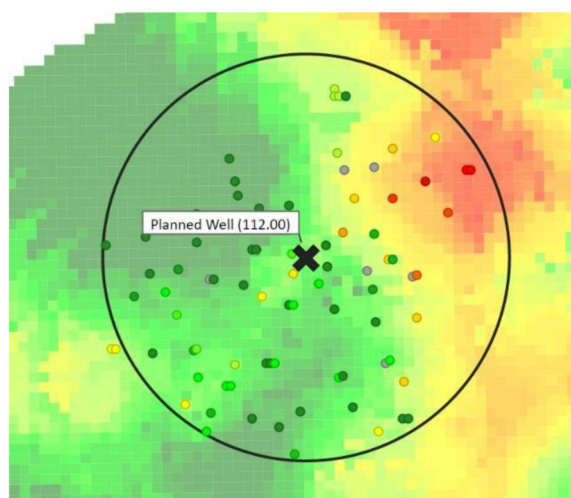


Figure 1—Offset wells plotted on heat map

Since this heat map combines a statistical analysis with mathematical computation, it yields an accurate estimate of drilling fluid density, which prevents flow while at the same time avoids encountering losses that can seriously hamper progress. In sensitive areas, the use of MPD becomes imperative, which has a significant cost impact. To overcome these challenges, the heat map becomes an essential tool that has rescued several wells and is continuously being refined as more data becomes available.

The analysis of offset wells with regard to encountered drilling troubles was done manually by the drilling engineering by reviewing the end-of-well-reports and the daily drilling reports. This could take several days, and eventually, driving a decision on the selected mud weight to drill the section would be a subjective call. The automation of this process using the latest available digital technologies is a landmark achievement that reduces this analysis to a few minutes and enables immediate decision-making thereby reducing an engineer's workload and freeing up resources to be applied elsewhere for improving the planning process.

Design Optimization

In the planning phase, the drilling team followed a bottom-up approach to well design, which was the key to optimizing the well design. Focusing on the end result, which would be a side-tracked or twinned well (slight nudge away from the mother bore), the following Geological requirements and constraints were identified for this project:

- Drill a vertical well to the deepest objective, setting a production casing above the flow zone and a production liner in the cap rock above the reservoir

- Run open-hole formation evaluation logs
- Based on logging results, the well must be twinned or side-tracked
- In the twinned or side-tracked lateral, run a production liner string to isolate the water zone
- Continue drilling into reservoir section and complete the well with conventional sand-screens completion

The conventional approach to such well construction projects has been to drill and set a production casing above the flow zone as explained before. Then, drilling continues below the flow zone, and a production liner is set to isolate it. Drilling further continues into the reservoir section, where logs are run and a decision is made to side-track or twin the well. [Figure 2](#) illustrates this conventional design configuration and the well construction phases. In certain instances, a well may be twinned below the shoe of the production liner in case a sufficient section of open hole is available below the shoe. However, in case this space is not available, then the only feasible option is to mill the production liner and re-drill the open hole below the production casing. In this case, once the well has been twinned or side-tracked, the production liner is run again to isolate the water zone existing just above the reservoir layer.

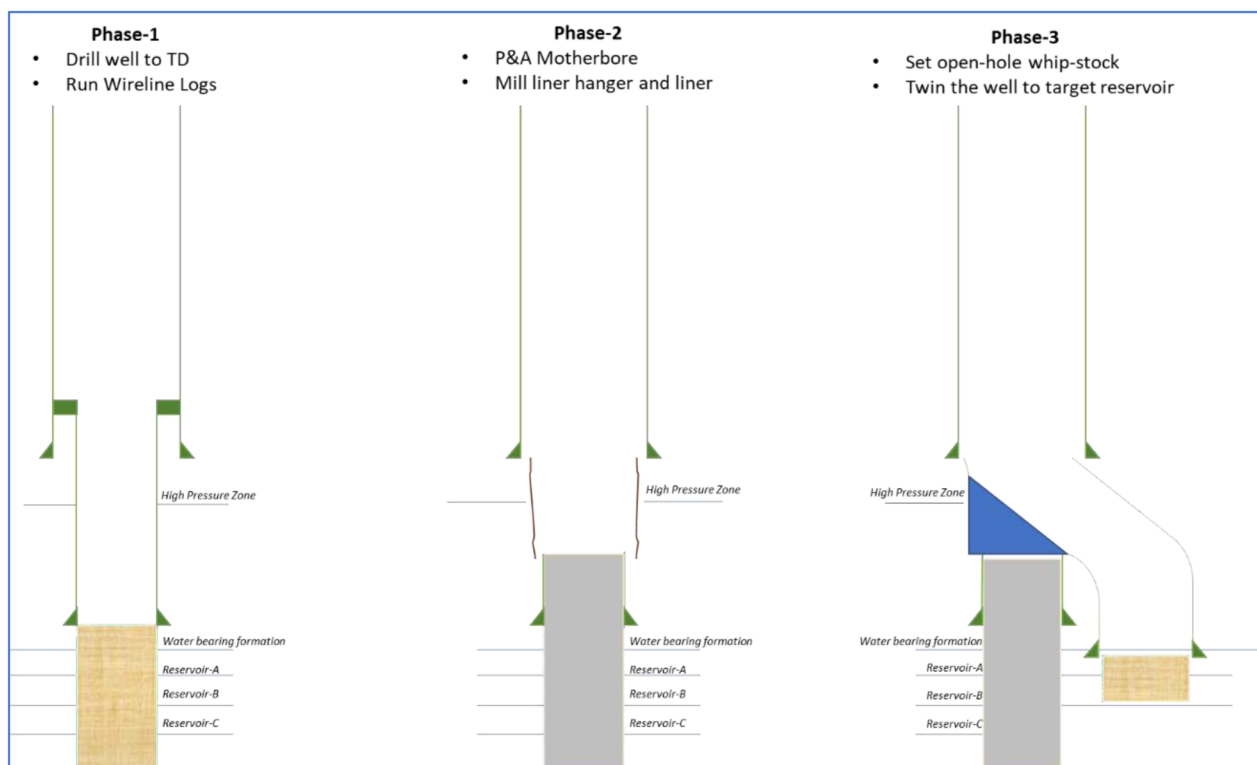


Figure 2—Conventional Well Design and Well Construction Phases

The sequence of operations in this case is as follows:

1. Drill a vertical well till above the flow zone and set production casing
2. Continue drilling through the flow zone with a high mud weight and set a production liner in the cap rock above the reservoir
3. Continue drilling into the reservoir formations
4. Run open-hole formation evaluation logs
5. Based on logging results, decide either to twin or side-track the well

6. In case of twinning, check if sufficient space is available below the production liner to nudge the well away and obtain sufficient departure from the mother-bore. If this is not possible or if the well is to be side-tracked, then proceed as follows:
 - a. Plug and abandon the mother-bore
 - b. Mill the production liner until 250 ft below the production casing
 - c. Set a whip-stock and initiate a side-track
 - d. Drill directionally as per plan
 - e. In the twinned or side-tracked lateral, run production liner string to isolate the water zone.
 - f. Continue drilling into the reservoir section and complete the well with conventional sand-screens completion

The improved well design is shown in [Figure 3](#). Keeping in mind that the side-tracking or twinning is a given, the drilling team focused on avoiding having to mill the production liner. This would only be possible if the production liner was not run, rather all the hole below the production casing was drilled in one section. However, this entailed a high risk of flow and, as a consequence, a risk of losses at kill mud weights. Here, the flow potential heat map became an enabler and predicted a mud weight that would not only allow drilling the well without incurring an influx but at the same time avoid losses in the formations below it as well as in the reservoir layers. Keeping in mind that one of the reservoir layers comprises unconsolidated sand that is prone to losses at or above a certain drilling fluid density, this tool came in handy to navigate through the imminent challenges.

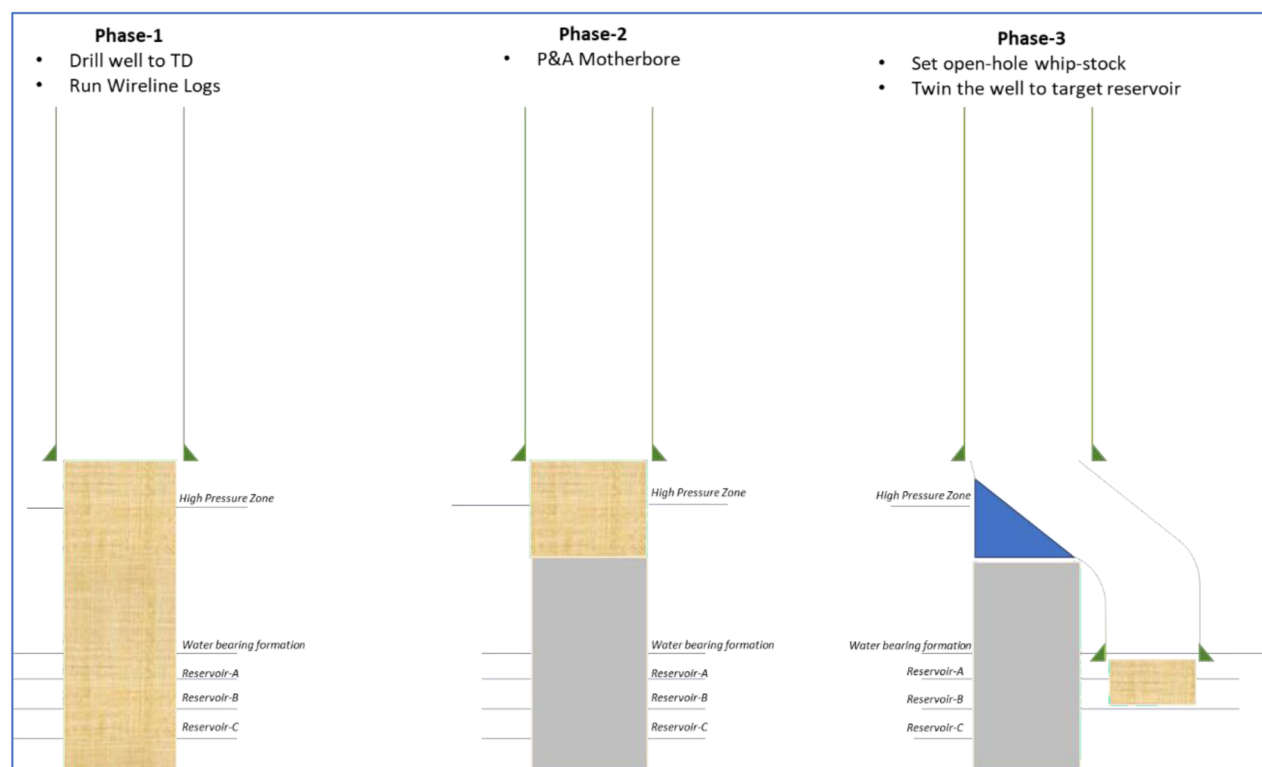


Figure 3—Improved Well Construction Process

By utilizing the heat map and designing a suitable mud weight, the Drilling and Gas Reservoir teams agreed to an improved design, which would be executed as follows:

- Drill a vertical well to the deepest objective, set only a production casing above the high-potential flow zone.
- Continue drilling the potential flow zone together with the cap rock and the reservoir section. As a result, the flow zone and reservoir layers would be in the same hole section
- Run open-hole formation evaluation logs
- Based on logging results, plug back the well in the open hole
- Set an open hole whip-stock, and twin or side-track the well
- In the twinned or side-tracked lateral, run a production liner string to isolate the water zone
- Continue drilling into reservoir section and complete the well with conventional sand-screens completion

By following a bottom-up approach, the drilling team was also able to optimize the well path. A suitable kick-off point was selected, which minimizes dog legs and leads to a smooth S-shape profile. Moreover, it provided flexibility to cater for both cases of twinning (short S-shape profile) as well as for a long S-shape side-track, the latter being possible to achieve from a shallower kick-off depth. The low dog leg and smooth profile for both scenarios took into account ease of running production liner or lower completion, and even catered for future well interventions or the possibility of running production evaluation logs during the life cycle of the well.

The drawback of this modification was that the rate of penetration (ROP) in the deeper formations would be relatively less, which would affect performance to some extent. It has been observed historically that drilling reservoir sections in a smaller hole size yields higher ROP with a smaller number of bit trips. However, the benefits envisioned by following this design far outweigh the reduction in drilling performance. Moreover, the well design would revert to the conventional scenario in case a kick was encountered from the flow potential zone. This would render the entire performance initiative useless. The upside in this case presented a tempting opportunity, and by adequately addressing and mitigating the identified risks, the team made an engineering decision to reap the maximum benefits.

By having these options available already in the planning phase, the drilling team developed a suitable Plug and Abandonment Program for the mother-bore and optimized the plan for open hole kick-off utilizing an open hole whip-stock. Having done all the planning upfront taking various factors in to consideration, the team was ready to deploy the design and push for performance.

Optimized Well Construction

Drilling activities commenced as per plan, and no major issues were encountered throughout the drilling process. After setting the production casing and drilling through the potential flow zone, the drilling team exercised an added precaution of performing extended flow checks in the open hole after drilling past the flow zone to ensure that the well does not flow. Drilling commenced through this zone without any issues and continued down to the deepest objective as per plan. Due to the correctly sized drilling fluid density, no well control scenario was encountered which proves the effectiveness of the heat map and mud weight prediction tool. The well was successfully logged and evaluated, after which the mother bore was plugged back and open hole twinning was performed with a whip- stock. The lateral was drilled to the top of the reservoir while carefully isolating a water zone right above it. Production liner was set, and then drilling continued into the reservoir as per plan. Eventually, the well was completed and delivered with minimal lost time while avoiding milling the liner top that would consume a considerable amount of time and carry a significant cost burden.

Conclusions

Utilizing a bottom-up approach to drilling plans, keeping in view the geological objectives, can help to streamline operations and prepare for the life-cycle of the well while yielding substantial time and cost savings. By combining hole sections, a high degree of flexibility can be achieved, especially when it is known that the pilot hole will be plugged and lateral will be drilled based on log evaluation. Knowing upfront which reservoir formations need to be targeted can help optimize the well path for ease of running tubulars and completion and future well interventions.

Developing in-house software for risk assessment of critical flow zones is a critical enabler when combining different zones in the wellbore. A thorough analysis of offset wells can provide valuable insights and help to mitigate the risk of influx by properly designing the drilling fluid density. Here, the utilization of latest available digital technologies can come in very handy for quickly analyzing offset wells, thereby enabling quick decisions. Large sets of available data can be quickly processed and displayed in a user-friendly interface with powerful visualizations. Such heatmaps can be utilized for other purposes, such as loss circulation risk assessment, stuck pipe prediction, well drifting tendency analysis, and can be combined together to provide a comprehensive analysis tool for the drilling engineer. Such digital technologies have a vast potential to bring operational efficiency, and time and cost saving.